Fourier Series

Course: Math 414

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Core Idea

We want to be able to use a combination of sin and cos to build up to any possible function

Definitions

For a function $f(x)$ on the interval $[-\pi, \pi]$, the fourier series is

$$a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) \tag{1}$$

Key Theorems

Theorem 1.1 The following integral relations hold.

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos(nx)\cos(kx)\,\mathrm{d}x = \begin{cases} 1 \text{ if } n = k \geq 1 \\ 2 \text{ if } n = k = 0 \\ 0 \text{ otherwise} \end{cases} \tag{2}$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin(nx)\sin(kx)\,\mathrm{d}x = \begin{cases} 1 \text{ if } n = k \geq 1 \\ 0 \text{ otherwise} \end{cases} \tag{3}$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos(nx)\sin(kx)\,\mathrm{d}x = 0 \quad \text{for all integers } n, k. \tag{4}$$

This means that the set of all $\left\{\frac{\cos(nx)}{\sqrt{\pi}}, \frac{\sin(nx)}{\sqrt{\pi}}\right\}$ is orthogonal.

Proof idea: Use the identities

$$\cos((n+k)x) = \cos(nx)\cos(kx) - \sin(nx)\sin(kx) \tag{5}$$

$$\cos((n-k)x) = \cos(nx)\cos(kx) + \sin(nx)\sin(kx) \tag{6}$$

Then either add or subtract them to get an equivalence for $\sin(nx)\sin(kx)$or $\cos(nx)\cos(kx)$.

For $\cos(nx)\sin(nx)$, observe it's an odd function multiplied by an even function

Theorem 1.2 Given the fourier series, we can find the coefficients by

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,\mathrm{d}x \tag{7}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x \tag{8}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,\mathrm{d}x \tag{9}$$

Proof idea: Multiply from the base Fourier series either $\cos(nx)$ or $\sin(nx)$ and integrate. Only the terms with

$$n = k \tag{10}$$

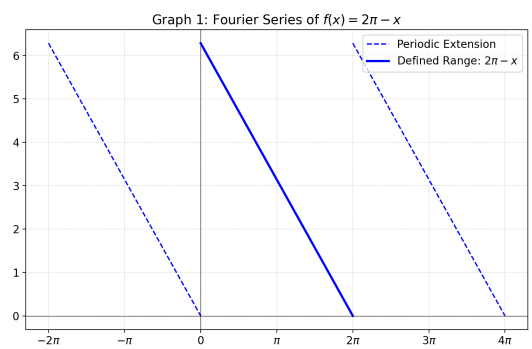
will contribute from above, which, giving us the above.

Lemma 1.3: The above only works for on the interval of $[-\pi, \pi]$, But this holds for any interval of length 2π according to this lemma: Assume f is 2π periodic and c is a real number:

$$\int_{-\pi+c}^{\pi+c} f(x)\,\mathrm{d}x = \int_{-\pi}^{\pi} f(x)\,\mathrm{d}x \tag{11}$$

Example

Come up with the fourier series for $f(x) = 2\pi - x$ on $[0, 2\pi]$



$$a_0 = \frac{1}{2\pi}\int_{0}^{2\pi} (2\pi - x)\,\mathrm{d}x \tag{12}$$

Using the graph, we can see this is just a triangle with base and height of 2π, plugging that in for the integral gives us:

$$a_0 = \frac{1}{2\pi}\frac{1}{2}(2\pi)^2 \tag{13}$$

$$a_0 = \frac{1}{2}2\pi \tag{14}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi}(2\pi - x)\cos(nx)\,\mathrm{d}x \tag{15}$$

Integrating by parts

$$a_n = \frac{1}{\pi}\left[(2\pi - x)\frac{1}{n}\sin(nx)\right]_0^{2\pi} - \int_0^{2\pi} -\sin(nx) \tag{16}$$

$$a_n = \frac{1}{\pi}\left(-\left[\frac{1}{n}\cos(nx)\right]_0^{2\pi}\right) \tag{17}$$

$$a_n = -\frac{1}{\pi n}(\cos(2\pi n) - \cos(0)) \tag{18}$$

$$a_n = -\frac{1}{\pi n}(1 - 1) = 0 \tag{19}$$

Now solving for b_n:

$$b_n = \frac{1}{\pi}\int_0^{2\pi}(2\pi - x)\sin(nx)\,\mathrm{d}x \tag{20}$$

Using integration by parts again with $u = 2\pi - x$ and $\mathrm{d}v = \sin(nx)\,\mathrm{d}x$:

$$b_n = \frac{1}{\pi}\left(\left[-\frac{2\pi - x}{n}\cos(nx)\right]_0^{2\pi} - \int_0^{2\pi}\left(-\left(-\frac{1}{n}\cos(nx)\right)\right)\mathrm{d}x\right) \tag{21}$$

$$b_n = \frac{1}{\pi}\left(\left[-\frac{2\pi - x}{n}\cos(nx)\right]_0^{2\pi} - \int_0^{2\pi}\frac{1}{n}\cos(nx)\,\mathrm{d}x\right) \tag{22}$$

The integral of cosine over a full period is zero, so the second term vanishes:

$$b_n = \frac{1}{\pi}\left[-\frac{2\pi - x}{n}\cos(nx)\right]_0^{2\pi} \tag{23}$$

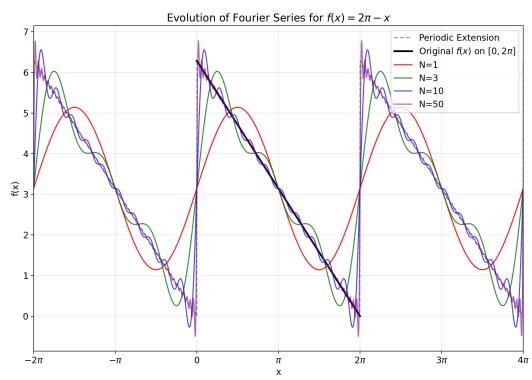
$$b_n = \frac{1}{\pi}\left(\left(-\frac{2\pi - 2\pi}{n}\cos(2\pi n)\right) - \left(-\frac{2\pi - 0}{n}\cos(0)\right)\right) \tag{24}$$

$$b_n = \frac{1}{\pi}\left(0 - \left(-\frac{2\pi}{n}\cdot 1\right)\right) \tag{25}$$

$$b_n = \frac{1}{\pi}\left(\frac{2\pi}{n}\right) \tag{26}$$

$$b_n = \frac{2}{n} \tag{27}$$

Putting it all together for the final Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos(nx) + b_n\sin(nx)) \tag{28}$$

$$f(x) = \pi + \sum_{n=1}^{\infty}\frac{2}{n}\sin(nx) \tag{29}$$